

Computer Networks Exercises

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Exercise 1

- Assuming a classless addressing plan, define the netmask and the longest prefix length that have to be assigned to possible networks in order to contain the given number of hosts.

Number of Hosts	Netmask	Prefix Length	Available Addresses
2			
27			
5			
100			
10			
300			
1010			
55			
167			
1540			



Exercise 2

- Which “IP address / Prefix length” pairs identify a valid network?

IP / Prefix Length pair	Valid network?
192.168.5.0/24	
192.168.4.23/24	
192.168.2.36/30	
192.168.2.36/29	
192.168.2.32/28	
192.168.2.32/27	
192.168.3.0/23	
192.168.2.0/31	
192.168.2.0/23	
192.168.16.0/21	
192.168.12.0/21	



Exercise 3

- Find out the configuration error in the network setup below and explain why the network does not work properly.

